Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Solving Systems by Graphing

Name: ______________
Date: __________

Foundation (Jalapeno)

Q1. What is the x-coordinate of the solution to the system $x + y = 2$ and $x - y = -2$?
(A) -2
(B) 0
(C) 2
(D) 4
(E) 6

Q2. The system $2x + 3y = 7$ and $x - 2y = -3$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 3
(E) Infinitely many

Q3. The point $(2, 3)$ is a solution to which system of equations?
(A) $x + y = 5$ and $x - y = -1$
(B) $x + y = 5$ and $x - y = 1$
(C) $x - y = 5$ and $x + y = -1$
(D) $x - y = 5$ and $x + y = 1$
(E) $x + y = -5$ and $x - y = -1$

Q4. The graphs of $y = 2x - 1$ and $y = -3x + 2$ intersect at the point with y-coordinate
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q5. What is the equation of the line passing through $(2, 3)$ and $(4, 5)$?
(A) $y = x + 1$
(B) $y = x - 1$
(C) $y = x + 2$
(D) $y = 2x - 1$
(E) $y = x - 2$

Q6. The system $y = 2x$ and $y = -2x$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 3
(E) Infinitely many

Q7. The equation $2x + 5y = 11$ can be written as $y =$
(A) $-$
rac25x +
rac115
(A)
rac25x +
rac115
(A) $-$
rac52x +
rac112
(A)
rac52x -
rac112
(A)
rac211x +
rac511

Q8. Which of the following systems has no solution?
(A) $x + y = 2$ and $x + y = 3$
(B) $x - y = 1$ and $x + y = 2$
(C) $y = 2x$ and $y = -2x$
(D) $y = x + 1$ and $y = x - 1$
(E) $x + y = 1$ and $x - y = 2$

Open Ended Questions (FRQ)

Q9. Solve the system of equations $x + 2y = 4$ and $3x - 2y = 5$ graphically. Find the solution and interpret it.

Q10. Consider the system $y = 2x - 1$ and $y = x + 2$. Graph both equations on the same coordinate plane, find the point of intersection, and interpret the result.

Chef's Tips

- Tip 1: Ensure students understand that graphical solutions involve finding the point(s) of intersection between two lines.
- Tip 2: Remind students to check their work by plugging the solution back into the original equations.
- Tip 3: Encourage students to use graph paper to accurately plot the lines.